CSC 251 HOMEWORK EIGHT

SPORTS CAR PRICE MODELS

THE ASSIGNMENT

This assignment provides practice in exploring data sets, merging data sets, performing simple and multiple regression analysis, and utilizing models.

PORSCHE AND JAGUAR DATA

You and your friend are debating how Porsche or Jaguar sports cars maintain their value as their age and mileage increase. You'd like to know whether the age or mileage of the car (or both) better determines the value of the car. You each gather some price data from the internet, you for a handful of Porsches, your friend for a handful of Jaguars.

The two datasets are called `Porsche.csv` and `Jaguar.csv`. Each file provides datapoints for 30 cars. The price is given in units of thousands of US dollars, age is given in years, and mileage is given in thousands of miles.

Load these datasets into a fresh notebook and explore them with the tools discussed in class. Combine the datasets into one DataFrame such that each Porsche and each Jaguar entry is an independent observation. Create simple and multiple linear regression models for each type of car and for the cars collectively. Use age and mileage as the predictor (independent) variables and price as your response (dependent) variable. There is no need to reserve a portion of the data for model validation.

You will specify predictive model equations for EACH brand of car and for all the cars collectively then provide the predicted prices for 10-year-old cars with 55,000 miles.

EXPECTATIONS

Be sure to include the following elements in your notebook:

- Your name, date, and a description of the lab in an opening Markdown cell. Include at least 2 different levels of Markdown headings.
- Installation of any packages your script uses in the first code cell. You should only use packages discussed in class which may include NumPy, pandas, matplotlib, Seaborn, statsmodels, and sklearn. Plotly might be helpful in your investigation but should not be included in your final notebook.
- Loading of the CSV data files into DataFrames and merging the resulting DataFrames within the notebook.

- Comments in Markdown cells in addition to in-code comments that provide a narrative of your notebook before each cell or group of cells. The Markdown headings specified below can be combined with your Markdown narrative to assist you in organizing your notebook.
- Exploration of the datasets and creation of several simple and multiple linear regression models.
- Graphical visualizations for all 6 of your single regression models: plot the 3 models using age (use Markdown heading **AGE_MODEL**) as the predictor variable against Porsches, Jaguars, and all cars on the same plot or as plots within the same figure. Likewise for the 3 models using mileage as the predictor (use Markdown heading **MILEAGE_MODEL**). Title and label your plots appropriately with variable names and units.
- **BONUS (+2 pts)**: One 3D plot for the best multiple regression model from your analysis of the three options (Porsche, Jaguar, and all cars). Title and label appropriately (use Markdown heading **3D_MODEL**).
- Code to compute the predictions for the price (USD) of a 10-year-old Porsche, Jaguar, and general sports car with 55,000 miles according to your multiple regression model equations. (3 different predictions) Complete the calculation in your notebook using script to access the intercept and slope values so that your notebook works with other datasets of the same format. Assign constants rather than use magic numbers and use Markdown headings **PREDICT_PORSCHE**, **PREDICT_JAGUAR**, and **PREDICT_ALL**.
- Answers to the following questions (with **Q1-Q4** headings indicated) into **separate** Markdown cells at the bottom of your notebook to answer the following questions:

  Q1: Is mileage or age or both factors more strongly correlated to the value of a Porsche? Of a Jaguar? Of all the sportscars combined?
  Q2: What is the predictive equation from your best model(s) for Porsches? For Jaguars? For all the cars combined? Write your equations in terms of years, miles, and US dollars, rounding to 2 decimal values, keeping in mind the units for the models. If two models are equally as strong, provide both equations.
  Q3: Summarize your predictions for the prices of a 10-year-old Porsche, a Jaguar, and a general sports car with 55,000 miles according to your multiple regression model equations to the nearest US dollar.
  Q4: How accurate and appropriate is the combined model with all the cars for Porsches? Jaguars? Provide reasoning for your answer.

Extraneous plots and analysis in your notebook should be deleted or commented out. Be sure to run your notebook from top to bottom again before saving.

SUBMISSION

Submit a single Jupyter notebook called *`HW8LastNameFirstInitial`*`.ipynb`. Run your notebook before saving so that it does not need to be run before grading.